



# Forecasting hospital drug demand for demand patterns with changepoints

Tianyi Xu<sup>a\*</sup>, Hayo Bos<sup>b,c\*</sup>, Eunshin Byon<sup>a</sup>, Mariel S. Lavieri<sup>a</sup>, Karl Renius<sup>d</sup>, and Burgunda V. Sweet<sup>e</sup>

<sup>a</sup>Department of Industrial and Operations Engineering, University of Michigan, Ann Arbor, MI, USA; <sup>b</sup>Diakonessenhuis Utrecht, Utrecht, The Netherlands; <sup>c</sup>Center for Healthcare Operations Improvement and Research, University of Twente, Enschede, The Netherlands; <sup>d</sup>Michigan Medicine, Department of Pharmacy Services, University of Michigan, Ann Arbor, MI, USA; <sup>e</sup>College of Pharmacy, University of Michigan, Ann Arbor, MI, USA

## ABSTRACT

Predicting drug demand is of great importance for hospital pharmacy managers. Drug expenditures per hospital in the US averages over 7 million dollars per year, so it is important to manage these costs effectively. Due to erratic demand patterns, however, predicting demand for drugs is not an easy task. One of the patterns that occur when predicting drug demand are demand patterns with changepoints. These non-stationary patterns suddenly change with respect to their mean, which might occur due to drug authority approvals or events such as a pandemic. In this paper, we study demand patterns with changepoints, which we found to affect 25.9% of the top 500 drugs at our partnering hospital. We propose to use a Bayesian Online Changepoint Detection model. The overall performance of this method is at least as good as other methods under regular circumstances, but outperforms other commonly used forecasting tools especially right after the changepoint occurs, showing its ability to quickly adapt to changepoints. Furthermore, changepoints are more likely to occur for pricier drugs. In particular, drugs within the third and fourth quantile price range are almost three times more likely to have a changepoint compared to inexpensive drugs in the first quantile. This implies that especially for pricier drugs, specialized changepoint models may be useful in predicting drug demand.

## ARTICLE HISTORY

Received 27 November 2024  
Accepted 18 July 2025

## KEYWORDS

Drug demand forecasting;  
pharmaceutical demand;  
Bayesian online  
changepoint detection;  
erratic demand;  
changepoints; predictive  
modeling; healthcare

## 1. Introduction

Drug demand in hospital pharmacy systems is an important but often overlooked facet of hospital capacity planning (Rachuba et al., 2023). In 2022, nonfederal hospitals in the US spent more than 37 billion dollars on prescription drugs (Tichy et al., 2023). Given that there are 5137 nonfederal hospitals (American Hospital Association, 2023), the drug expenditure per nonfederal hospital was more than 7 million dollars. Improper inventory management causes shortages and wastage, which, in the former case leads to lower quality of care, and in the latter case to a waste of money. Hence, careful drug inventory planning is of utmost importance. An essential factor in inventory planning is a reliable estimate of the demand in advance, which needs to be met with the inventory at hand.

The key element of a hospital pharmacy system is the Central Pharmacy (CP), which orders from wholesalers and supplies satellite stock locations in the hospital. Most of the hospital's drug inventory is kept at the CP. Orders are placed following an inventory policy. A first step to improve the inventory policy is an accurate demand forecasting mechanism (Silver et al., 2017, Ch. 3). Drug demand is often

subject to uncertainty, making accurate forecasting more challenging. As pointed out in Gebicki et al. (2014), however, waste and stock-outs are likely to decrease if drug demand forecasts are more accurate.

Drug demand is hard to predict, as the underlying patterns are often highly volatile. Drug demand patterns are shown to suffer from irregular or sporadic demand (Thaiudomsap & Sribundit, 2022). Additional challenging patterns include those with changepoints, where demand patterns show sudden shifts in their mean or variance. Figure 1 displays two examples of drugs with changepoints. In Fig. 1a, the normalized demand of the drug (003404) is around 0.4 units during the first 40 wk, but abruptly changes to around 0.1 units for the remaining 20 wk. In Fig. 1b, the mean of the normalized demand for the drug (003207) is around 0.2 units for the first two thirds of the periods, and it shifts to around 0.6 units for the remaining one third of the periods.

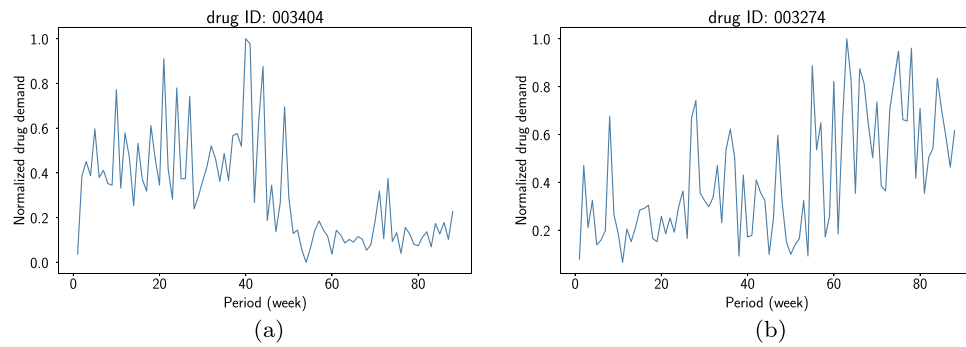
These changepoints could be driven by drug shortages, which in turn lead to an increase in demand of similar drugs (Shukar et al., 2021). Other drivers for changepoints could include the approval of the usage of a certain drug for

**CONTACT** Hayo Bos  [h.bos-1@utwente.nl](mailto:h.bos-1@utwente.nl)  Diakonessenhuis Utrecht, P.O. Box 80250, Utrecht, The Netherlands; Tianyi Xu  [tianyx@umich.edu](mailto:tianyx@umich.edu)  
 Department of Industrial and Operations Engineering, University of Michigan, Ann Arbor, MI, 48109, USA.

\*These authors contributed equally to this work.

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**Figure 1.** Examples of standardized drug demand time series that experience sudden changes.

a new group of diseases, or events such as a pandemic. In the US, approval can be changed at a national level, or a local policy level as well. For example, a medical center could restrict or expand formulary usage of particular products for a variety of reasons (cost, efficacy, etc.). Changes in demand due to policy changes are most likely to be due to the local variety. Forecasting is especially hard shortly after changepoints occur. Despite the occurrence and impact of changepoints in drug demand in hospitals, to the best of our knowledge, we are not aware of work focusing on this type of demand patterns in a hospital setting.

A major challenge in drug demand forecasting lies in the vast number of drug types that hospital inventories must manage, making it impractical to manually assess potential demand shifts for each drug. Without timely detection, inventory management teams may only realize demand changes after adverse consequences, like stock-outs or rising costs, begin to appear. To address this, an automated surveillance model that monitors all drugs and flags those experiencing potential demand shifts is essential. Such a model could improve demand predictions, enabling proactive inventory management and reducing the likelihood of supply disruptions.

In this paper, we present a Bayesian Online Changepoint Detection (BOCD) model to cope with changepoints in the demand for hospital drugs. This model detects changepoints on the fly, and it is updated with each new datum that becomes available over time. The model automatically detects changepoints, naturally providing better predictions for those drugs, i.e., predictions with smaller prediction errors, particularly close to the changepoints. We show that we do not compromise on the predictive power of this method for drugs without changepoints compared to other frequently used methods. The model could therefore save valuable time that inventory managers need to react to demand shifts.

We summarize our contributions as follows. First, to the best of our knowledge, this is the first study to address the drug demand forecasting by accounting for potential abrupt changes due to unknown factors. Our procedure characterizes uncertainties in changepoint occurrences and provides robust forecasting capability. Second, we extend the analysis from single-drug forecasting to multiple-drug cases. Our case study demonstrates that multivariate drug analysis is effective in identifying changepoints and reducing false

alarms for drugs with highly correlated demand patterns. Lastly, probabilistic forecasting results can be leveraged to optimize inventory management, aiming to minimize the combined costs of understocking and overstocking. Our case study demonstrates that our approach could be valuable for achieving robust inventory management.

The remainder of this paper is organized as follows. In Section 2, we present related literature on the broader topic of drug demand forecasting in hospitals. Section 3 presents our prediction model, which we apply on real-world data in Section 4. Section 5 presents the practical applications of our proposed method for inventory management. We reflect upon our work in Section 6, in which we also provide possible avenues for further research.

## 2. Related work

There is a substantial volume of literature about forecasting. Scoped to healthcare logistics, forecasting methods find extensive applications. Non-model based approaches for time series forecasting include moving average, exponential smoothing, and Holt-Winters. Despite their interpretability, these methods face occasional criticism for sub-optimal performance, as indicated in Whitt and Zhang (2019), or inconsistent outcomes, as discussed in Jones et al. (2008).

Model based time series forecasting techniques include Auto Regressive (AR) models and their extensions. In these models, past observations serve as regressors. There are plenty of applications in healthcare logistics. For instance, Jones et al. (2009) introduce a Vector Auto Regressive (VAR) model to predict Emergency Department (ED) demand and demand for subsequent resources simultaneously. Combining AR and Moving Average (MA) results in Auto Regressive Moving Average (ARMA) and Auto Regressive Integrated Moving Average (ARIMA) models, tailored for stationary and non-stationary time series, respectively. These models have found application in forecasting ED demand (Carvalho-Silva et al., 2018), pediatric care demand (Kadri et al., 2014), and outpatient visit demand (Huang et al., 2020).

Incorporating seasonality and exogenous variables gives rise to Seasonal Auto Regressive Integrated Moving Average (SARIMA) and Seasonal Auto Regressive Integrated Moving Average Exogenous (SARIMAX) models, respectively. For example, Jones et al. (2008) explore a SARIMA model to

predict ED demand, while Zinouri et al. (2018) employed it to forecast surgical case demand. Moreover, Whitt and Zhang (2019) consider a SARIMAX model for ED demand prediction, finding that weather and calendar variables make their prediction more accurate.

Recent trends reveal the growing popularity of machine learning methods in healthcare logistics. In Dunstan et al. (2023), pediatric no-shows are predicted using machine learning techniques like Random Forest, Logistic Regression, and Support Vector Machines. Comparably, Alaeddini et al. (2015) predict no-shows at outpatient clinics using Logistic Regression and Bayesian inference, and Erekat et al. (2020) predict surgery cancellations using an ensemble model. Arora et al. (2023) predict patient waiting times in an ED using quantile regression forests. Similarly, Suleiman et al. (2022) employ Bayesian quantile regression for ED waiting time prediction. Other applications include drug shortage prediction (Pall et al., 2023), surgical length-of-stay prediction (Shi et al., 2023) and emergency spatiotemporal call volume prediction (Martin et al., 2021).

Despite the rise of machine learning, there is no unanimous agreement on whether machine learning methods or classical time series models perform better. Comparative studies, such as those in Whitt and Zhang (2019); Jones et al. (2008), suggest that the choice depends on the specific context. They favor classical models over machine learning models in some cases. On the contrary, Schäfer et al. (2023) integrate machine learning predictions into solving the Patient Bed Assignment problem, demonstrating the superiority of machine learning methods over classical time series methods in that context.

A notable limitation of machine learning methods is their tendency to provide point estimates only. To address this, techniques like Quantile Regression Forests, as demonstrated in Dean et al. (2022), can be used. Alternatively, Bayesian methods, as applied in Suleiman et al. (2022), offer the complete posterior distribution.

Drug demand forecasting has been done for various applications, including retail pharmacies (Burinskiene, 2022; Papanagnou & Matthews-Amune, 2018), aggregated supply chains (Merkuryeva et al., 2019; Pall et al., 2023; Yani & Aamer, 2023), and hospital pharmacies (Luochen & Hasachoo, 2021; Thaiudomsap & Sribundit, 2022). Traditional demand forecasting techniques are used in Burinskiene (2022), Luochen and Hasachoo (2021), Merkuryeva et al. (2019), and Thaiudomsap and Sribundit (2022). In Papanagnou and Matthews-Amune (2018), a SARIMAX model using Google hits for specific drug related terms is proposed. More recently, machine learning has gained popularity in pharmaceutical demand forecasting (Pall et al., 2023; Yani & Aamer, 2023).

Demand forecasts in the pharmaceutical context are mostly used to optimize stock levels and hedge against fluctuations, such as in epidemics (Paul & Venkateswaran, 2018). Examples of previous work on inventory optimization in the context of pharmacies include Gebicki et al. (2014) who use simulation, Guerrero et al. (2013) who use mathematical models and Rosales et al. (2015) and Saha and Ray (2019) who use Markov models. However, as noted in

Goltsos et al. (2022), forecasting demand and optimizing inventory decisions are often separate worlds. Conditions for the collaboration of inventory systems of hospital pharmacies are explored in Czerniak et al. (2024b).

As noted in the introduction, one of the key challenges of forecasting drug demand compared to other areas is the wide variety of demand patterns, due to, e.g., supply issues, changing regulations, seasonality, and epidemics (Shukar et al., 2021). This should be taken into account when forecasting future demand, for instance, by clustering similar demand patterns as done in Zhu et al. (2021). After clustering, methods per pattern type can be considered such as in (Czerniak et al., 2024a; Thaiudomsap & Sribundit, 2022), where particular emphasis is given to irregular drug demand. The latter work addresses inventory policies in the context of irregular demand. Despite their importance in the pharmaceutical context, forecasting demand patterns with changepoints are not yet addressed in the pharmaceutical demand forecasting literature.

We conclude that existing forecasting studies do not directly model drug demand patterns with changepoints. Most work focuses on forecasting methods for general demand patterns, which might cause drug shortages and inefficiencies. However, specialized methods for drugs with changepoints might bear fruit as it is likely that quick adaptation to the situation after the changepoint will decrease potential waste.

On the other hand, most changepoint detection studies focus on pinpointing the occurrence of changepoints without providing subsequent forecasts (Aminikhanghahi & Cook, 2017). Saatçi et al. (2010) propose a changepoint model that integrates Gaussian Processes (GP) into the BOCD framework. However, their method suffers from increasing computational complexity over time, with the computational complexity of  $\mathcal{O}(T^4)$ , where  $T$  is the number of observations accumulated up to the current time.

In this study, we address this research gap by employing BOCD with a normal distribution. Our approach maintains  $\mathcal{O}(T^2)$  complexity and supports closed-form model updates, indicating its scalability for practical implementation, particularly in large inventory systems. Additionally, we extend the analysis from single-drug to multiple-drug forecasting by taking into account the correlation among drug demands. For the multivariate drug demand forecasting, our approach keeps  $\mathcal{O}(D^2T^2)$  computational complexity, where  $D$  is the number of drugs.

### 3. Methodology

In our study, we utilize the BOCD technique, first suggested by Adams and MacKay (2007), to forecast drug demand. This method enables us to make sequential predictions about future demand and at the same time, it allows us to identify changepoints-specific points in time when the underlying demand process undergoes significant changes. We first consider a single drug demand problem in Sections 3.1 to 3.3. Specifically, we formulate a changepoint model that leverages historical data in Section 3.1. In Section 3.2,

**Table 1.** Notation and symbols.

|                                         |                                                                                              |
|-----------------------------------------|----------------------------------------------------------------------------------------------|
| Index:                                  |                                                                                              |
| $t$                                     | time point                                                                                   |
| $i$                                     | drug index                                                                                   |
| Random variables for univariate data:   |                                                                                              |
| $x_t$                                   | drug demand at time $t$                                                                      |
| $x_{a:b}$                               | a sequence of observations from time $a$ to time $b$ , including $a$ and $b$                 |
| $r_t$                                   | run length at time $t$                                                                       |
| $\mu$                                   | demand mean                                                                                  |
| Parameters for univariate data:         |                                                                                              |
| $\lambda$                               | expected number of time points between two consecutive changepoints                          |
| $\sigma_x^2$                            | variance of the data                                                                         |
| $\mu_0$                                 | prior mean for $\mu$                                                                         |
| $\sigma_0^2$                            | prior variance for $\mu$                                                                     |
| $\mu_t^{(r)}$                           | posterior mean for $\mu$ given the run length at time $t$ is $r$                             |
| $\tau_t^{2(r)}$                         | posterior variance for $\mu$ given the run length at time $t$ is $r$                         |
| $\alpha_t^{2(r)}$                       | posterior precision for $\mu$ given the run length at time $t$ is $r$                        |
| Random variables for multivariate data: |                                                                                              |
| $x_t^{(i)}$                             | demand of the $i$ th drug at time $t$                                                        |
| $x_{a:b}^{(i)}$                         | demand sequence of the $i$ th drug from time $a$ to time $b$ , including $a$ and $b$         |
| $X_t$                                   | a vector of multiple drug demands at time $t$                                                |
| $X_{a:b}$                               | a matrix of multiple drug demand sequences from time $a$ to time $b$ , including $a$ and $b$ |
| $r_t^{(i)}$                             | marginal run length for the $i$ th drug at time $t$                                          |
| $\theta$                                | mean vector of multiple drug demands                                                         |
| Parameters for multivariate data:       |                                                                                              |
| $\theta_0$                              | prior mean vector for $\theta$                                                               |
| $\Sigma_0$                              | prior covariance matrix for $\theta$                                                         |
| $\theta_t$                              | posterior mean vector for $\theta$ given the entire historical data                          |
| $\Sigma_t$                              | posterior covariance matrix for $\theta$ given the entire historical data                    |
| $\theta_{t,r}^{(i)}$                    | conditional posterior mean for the $i$ th drug given the marginal run length $r$             |
| $\Sigma_{t,r}^{(i)}$                    | conditional posterior variance for the $i$ th drug given the marginal run length $r$         |

we present the method for updating the predictive distribution sequentially as new data points are observed, taking into account the potential occurrence of a changepoint. Section 3.3 discusses the process of identifying changepoints within the data. Section 3.4 extends the analysis to encompass multiple drug demand forecasting and changepoint identification. Table 1 presents a summary of the notations used for modeling weekly drug demand in both univariate and multivariate analyses.

### 3.1. Bayesian online changepoint detection for drug demand forecasting

Let  $x_t$  denote drug demand for a certain drug at time  $t$ . We assume that drug demand follows a probability distribution  $D$  with its parameters  $\theta$ . For instance, this could be a normal distribution with a mean  $\mu$  and variance  $\sigma^2$  over a particular period of time. At some point, a sudden change occurs in  $\theta$ . Following the change point, the demand sequence is generated from a new distribution with the updated  $\theta$  parameter. Therefore, the drug demand time series data  $x_1, \dots, x_t$  can be partitioned into several distinct, non-overlapping stationary segments. Such segments are

unknown a priori and must be inferred from the data. Changepoints are the specific moments when  $\theta$  undergoes these shifts. The underlying demand process with changepoints can be viewed as non-stationary, as the parameters across different segments vary. Specifically, a changepoint process can be treated as piecewise stationary, which is considered a special type of nonstationary time series.

A crucial step in identifying changepoints is to introduce the concept of *run length*. The run length, denoted as  $r_t$ , represents the number of time steps between the current time  $t$  and the most recent changepoint. It takes discrete values from 0 to  $t$ : 0, 1, 2, ...,  $t$ . Since we do not know the run length a priori, we treat it as a random variable and learn its distribution from the data.

Let  $x_{1:t}$  denote the data sequence we have observed up to time  $t$ , i.e.,  $x_{1:t} = \{x_1, x_2, \dots, x_t\}$ . Given the uncertainty of the run length, we formulate the joint distribution of the run length and the observed data using the forward algorithm (Adams and MacKay, 2007):

$$P(r_t, x_{1:t}) = \sum_{r_{t-1}} P(r_t | r_{t-1}) P(x_t | r_{t-1}, x_{1:t-1}) P(r_{t-1}, x_{1:t-1}). \quad (1)$$

Below, we provide a detailed explanation of each term on the right-hand side in (1). The first term  $P(r_t | r_{t-1})$ , represents the conditional changepoint prior. It can only take on two values: 0 and  $r_{t-1} + 1$ . When  $r_t = 0$ , it means that a changepoint occurs at time  $t$ . If a changepoint does not occur,  $r_t$  is incremented by one from its previous value, becoming  $r_{t-1} + 1$ . When information about past changepoints in drug demand is not available, we can employ a simple prior by assuming that the probability of a changepoint occurrence remains constant over time. Specifically, we have

$$P(r_t | r_{t-1}) = \begin{cases} \frac{1}{\lambda} & \text{if } r_t = 0, \\ 1 - \frac{1}{\lambda} & \text{if } r_t = r_{t-1} + 1, \\ 0 & \text{otherwise,} \end{cases} \quad (2)$$

where  $\lambda$  denotes the expected number of time points between two consecutive changepoints.

The second term in Equation (1),  $P(x_t | r_{t-1}, x_{1:t-1})$ , represents the conditional predictive distribution given all data since the last changepoint. This implies that if a changepoint happens, we build a new prediction model using only the data collected after the changepoint, rather than relying on the entire historical data up to the current period. Specifically, this involves utilizing the demand data between time  $t - r_{t-1}$  and  $t - 1$ . However, if a changepoint just occurs and no data is observed, we will build the predictive model based on the prior distribution. The exact form of this conditional predictive distribution will be presented in the next section.

The third term in Equation (1),  $P(r_{t-1}, x_{1:t-1})$ , is the recursive term obtained at time  $t - 1$ . In order to run this recursive algorithm, we must define the initialization condition  $P(r_0)$ . We consider the case where a changepoint is presumed to occur prior to the observation of the initial data



point. This suggests that we assign all of the probability mass for the initial run length at zero, i.e.  $P(r_0 = 0) = 1$ .

Equation (2) represents the prior belief of the run length distribution. We update this prior belief using the drug demand data to obtain the posterior run length distribution as follows,

$$P(r_t | x_{1:t}) = \frac{P(r_t, x_{1:t})}{\sum_{r_t} P(r_t, x_{1:t})}. \quad (3)$$

Finally, we obtain the predictive distribution by taking the weighted sum of the conditional predictive distributions over the posterior run length distribution as follows,

$$P(x_{t+1} | x_{1:t}) = \sum_{r_t} P(r_t | x_{1:t}) P(x_{t+1} | r_t, x_{1:t}). \quad (4)$$

Here, the last term uses the conditional distribution, given data  $x_{(t+1-r_t):t}$  collected after the last change point when the run length  $r_t$  is nonzero. If the run length is zero, the conditional distribution is constructed upon Bayesian prior, which will be discussed in the next section.

### 3.2. Bayesian predictive model

This section presents the predictive model, updated after new data is revealed. In Section 3.2.1 we discuss the conditional predictive model. In Section 3.2.2 we present the updating rules, and we give the posterior predictive model in Section 3.2.3.

#### 3.2.1. Conditional predictive model

Let  $X$  be a random variable representing the demand of a particular drug. We model  $X$  using a normal distribution with a normal prior on the mean and a fixed variance, that is,  $X \sim N(\mu, \sigma_x^2)$ . If the data is extremely skewed from a normal distribution, we can first apply a Box-Cox power transformation to the original data so that the transformed data becomes normally distributed (Box & Cox, 1964).

In this model, the changepoint refers to the time point where the mean parameter  $\mu$  changes. Within the Bayesian inference framework, we set the prior for  $\mu$  as  $\mu \sim N(\mu_0, \sigma_0^2)$ . When no data is observed, the predictive distribution for  $X$  take the prior distribution:

$$P(X | \mu_0, \sigma_0^2, \sigma_x^2) = \int_{\mu} P(X | \mu, \sigma_x^2) P(\mu | \mu_0, \sigma_0^2) d\mu. \quad (5)$$

This distribution takes a closed-form expression under normality assumptions as follows:

$$X | (\mu_0, \sigma_0^2, \sigma_x^2) \sim N(\mu_0, \sigma_0^2 + \sigma_x^2). \quad (6)$$

We now introduce how to compute the conditional predictive distribution  $P(x_t | r_{t-1}, x_{1:t-1})$  in Equation (1). If  $t = 0$  or  $r_{t-1} = 0$ , implying that a changepoint occurs just before time  $t$  and the process restarts, then the conditional predictive distribution is the prior predictive distribution. If  $r_{t-1} > 0$ , this suggests that we have observed some data since the last changepoint, i.e.,  $x_{(t-r_{t-1}): (t-1)}$ . Given that the drug demand between two consecutive changepoints is stationary,

$x_{(t-r_{t-1}): (t-1)}$  should be independently and identically distributed as  $N(\mu, \sigma_x^2)$ . The likelihood function is then given by

$$P(x_{(t-r_{t-1}): (t-1)} | \mu, \sigma_x^2) = \prod_{i=t-r_{t-1}}^{t-1} \frac{1}{\sqrt{2\pi\sigma_x^2}} \exp \left\{ -\frac{1}{2\sigma_x^2} (x_i - \mu)^2 \right\}. \quad (7)$$

Applying Bayes' rule, we update the distribution for  $\mu$  to get its posterior distribution, which depends on the run length,  $r_{t-1}$ , as follows,

$$\begin{aligned} P(\mu | x_{(t-r_{t-1}): (t-1)}) &\propto P(x_{(t-r_{t-1}): (t-1)} | \mu, \sigma_x^2) P(\mu) \\ &\propto \exp \left\{ -\frac{1}{2} \left[ \frac{\sum_{i=t-r_{t-1}}^{t-1} (x_i - \mu)^2}{\sigma_x^2} + \frac{(\mu - \mu_0)^2}{\sigma_0^2} \right] \right\} \\ &\propto \exp \left\{ -\frac{(\mu - \mu_t^{(r_{t-1})})^2}{2\tau_t^2(r_{t-1})} \right\}. \end{aligned} \quad (8)$$

It implies that the posterior distribution for  $\mu$  is normal:

$$\mu | x_{(t-r_{t-1}): (t-1)} \sim N(\mu_t^{(r_{t-1})}, \tau_t^2(r_{t-1})), \quad (9)$$

with

$$\mu_t^{(r_{t-1})} = \frac{\sigma_x^2 \mu_0 + \sigma_0^2 \sum_{i=t-r_{t-1}}^{t-1} x_i}{r_{t-1} \sigma_0^2 + \sigma_x^2}, \quad (10)$$

and

$$\tau_t^2(r_{t-1}) = \frac{1}{\frac{1}{\sigma_0^2} + \frac{r_{t-1}}{\sigma_x^2}}. \quad (11)$$

Next, the conditional posterior predictive distribution is calculated using

$$\begin{aligned} P(x_t | r_{t-1}, x_{1:t-1}) &= \int_{\mu} P(x_t | \mu) P(\mu | x_{(t-r_{t-1}): (t-1)}) d\mu \\ &\propto \int_{\mu} \exp \left\{ -\frac{1}{2\sigma_x^2} (x_t - \mu)^2 \right\} \\ &\quad \exp \left\{ -\frac{1}{2\tau_t^2(r_{t-1})} (\mu - \mu_t^{(r_{t-1})})^2 \right\} d\mu, \end{aligned} \quad (12)$$

which also has a closed-form expression as

$$x_t | r_{t-1}, x_{1:t-1} \sim N(\mu_t^{(r_{t-1})}, \tau_t^2(r_{t-1}) + \sigma_x^2). \quad (13)$$

#### 3.2.2. Sequential update rules

We update the model hyperparameters each time a new datum is observed, facilitating online learning. From Equations (10) and (11), at time  $t$ , for each value of  $r_{t-1}$ , we need to update  $\mu_t^{(r_{t-1})}$  and  $\tau_t^2(r_{t-1})$  using all data since the last changepoints. As time grows, the possible number of run length increases, and the number of hyperparameters to update also increases. Since the values of  $r_t$  depend only on  $r_{t-1}$ , we can rewrite the hyperparameters in a recursive manner to reduce computational efforts. Specifically, we want

$$\mu_{t+1}^{(r_t)} = f_1(\mu_t^{(r_{t-1})}, x_t), \text{ and} \quad (14)$$

$$\tau_{t+1}^2(r_t) = f_2(\tau_t^2(r_{t-1}), x_t), \quad (15)$$

for  $t = 0, \dots, t$ , where  $f_1(\cdot)$  and  $f_2(\cdot)$  are functions which are independent of past information  $x_{1:t-1}$ .

Our goal is to derive the exact forms for  $f_1(\cdot)$  and  $f_2(\cdot)$ . Conditional on  $r_{t-1}$ ,  $r_t$  can only take two values: 0 or  $r_{t-1} + 1$ . When  $r_t = 0$ , we know from Equation (6) that  $\mu_{t+1}^{(0)} = \mu_0$  and  $\tau_{t+1}^{2(0)} = \sigma_0^2 + \sigma_x^2$ . When  $r_t = r_{t-1} + 1$ , this implies that we observe a new datum  $x_t$  from the same distribution as the historical sequence  $x_{t-r_{t-1}:t-1}$ , so we can directly apply the rule for updating the posterior distribution in Bayesian inference.

Particularly, we first derive the recursive formula for  $\tau_{t+1}^{2(r_t)}$ . Let  $\alpha_t^{(r_t)} = \frac{1}{\tau_t^{2(r_t)}}$ , then we get

$$\alpha_{t+1}^{(r_t)} = \frac{1}{\tau_{t+1}^{2(r_{t-1}+1)}} = \frac{1}{\sigma_0^2} + \frac{r_{t-1} + 1}{\sigma_x^2} = \alpha_t^{(r_{t-1})} + \frac{1}{\sigma_x^2}. \quad (16)$$

Substitute  $\tau_t^{2(r_{t-1})}$  back and after rearranging the equation, we get

$$\tau_{t+1}^{2(r_t)} = \frac{1}{\frac{1}{\tau_t^{2(r_{t-1})}} + \frac{1}{\sigma_x^2}}. \quad (17)$$

Based on Equation (17), we rewrite  $\mu_{t+1}^{(r_t)}$  as

$$\begin{aligned} \mu_{t+1}^{(r_t)} &= \frac{\sigma_x^2 \mu_0 + \sigma_0^2 \sum_{i=t-r_{t-1}+1}^t x_i}{(r_t + 1)\sigma_0^2 + \sigma_x^2} \\ &= \frac{\sum_{i=t-r_{t-1}+1}^t x_i}{\frac{\sigma_x^2}{\sigma_0^2} + \frac{\mu_0}{\sigma_0^2} + \frac{x_t}{\sigma_x^2}} \\ &= \frac{\frac{r_{t-1}+1}{\sigma_x^2} + \frac{1}{\sigma_0^2}}{\left(\frac{\sum_{i=t-r_{t-1}+1}^t x_i}{\sigma_x^2} + \frac{\mu_0}{\sigma_0^2}\right) + \frac{x_t}{\sigma_x^2}} \\ &= \tau_{t+1}^{2(r_t)} \left( \frac{\sum_{i=t-r_{t-1}+1}^t x_i}{\sigma_x^2} + \frac{\mu_0}{\sigma_0^2} \right) + \tau_{t+1}^{2(r_{t-1}+1)} \left( \frac{x_t}{\sigma_x^2} \right) \\ &= \tau_{t+1}^{2(r_t)} (\alpha_t^{(r_{t-1})} \mu_t^{(r_{t-1})}) + \tau_{t+1}^{2(r_{t-1}+1)} \left( \frac{x_t}{\sigma_x^2} \right) \\ &= \frac{\tau_{t+1}^{2(r_t)}}{\tau_t^{2(r_{t-1})}} \mu_t^{(r_{t-1})} + \frac{\tau_{t+1}^{2(r_t)}}{\sigma_x^2} x_t. \end{aligned} \quad (18)$$

Overall, the sequential update rules become

$$\tau_{t+1}^{2(r_t)} = \begin{cases} \sigma_0^2 + \sigma_x^2 & \text{if } r_t = 0, \\ \frac{1}{\frac{1}{\tau_t^{2(r_{t-1})}} + \frac{1}{\sigma_x^2}} & \text{if } r_t = r_{t-1} + 1, \end{cases} \quad (19)$$

and

$$\mu_{t+1}^{(r_t)} = \begin{cases} \mu_0 & \text{if } r_t = 0, \\ \frac{\tau_{t+1}^{2(r_t)}}{\tau_t^{2(r_{t-1})}} \mu_t^{(r_{t-1})} + \frac{\tau_{t+1}^{2(r_t)}}{\sigma_x^2} x_t & \text{if } r_t = r_{t-1} + 1. \end{cases} \quad (20)$$

Regarding the computational complexity, at each time  $t$ , the cost to update the hyperparameters is  $\mathcal{O}(t)$ , as we need to update for each value of  $r_t$ , where  $r_t$  can take values from 0 to  $t$ . Consequently, the total cost up to time  $t$  is  $\sum_{i=1}^t \mathcal{O}(i) = \mathcal{O}(t^2)$ .

### 3.2.3. Posterior predictive model

Finally, with the posterior distribution of the mean parameter, the posterior predictive distribution  $P(x_{t+1}|x_{1:t})$  for the

drug demand at time  $t + 1$ , given the observations up to time  $t$ , becomes a Gaussian mixture distribution. In this distribution, the weight corresponds to the posterior run length probability  $P(r_t|x_{1:t})$ . In our analysis, we use the following mean of this distribution as the point prediction of the model:

$$\mu_{t+1} = \sum_{r_t} P(r_t|x_{1:t}) \mu_{t+1}^{(r_t)}. \quad (21)$$

The full BOCD procedure for online learning in combination with changepoint detection is presented in Algorithm 1. For uncertainty quantification, one can run simulations to obtain the full model distribution.

---

#### Algorithm 1. Drug demand forecasting with changepoint detection

---

- 1: **Input** : priors for  $\lambda$ ,  $\mu_0$ ,  $\sigma_0^2$
  - 2: Observe new datum  $x_t$
  - 3: Compute conditional predictive probability  $P(x_t|r_{t-1}, x_{1:t-1})$  using Equation (13)
  - 4: Compute growth probability  $P(r_t = r_{t-1} + 1, x_{1:t})$  using Equation (1)
  - 5: Compute changepoint probability  $P(r_t = 0, x_{1:t})$  using Equation (1)
  - 6: Compute posterior run length distribution  $P(r_t|x_{1:t})$  using Equation (3)
  - 7: Compute most probable run length  $\hat{r}_t$  using Equation (22)
  - 8: **Claim a changepoint** if  $\hat{r}_t < \hat{r}_{t-1} + 1$
  - 9: Update hyperparameters  $\mu_{t+1}^{(r_t)}, \tau_{t+1}^{2(r_t)}$  using Equation (19) and (20)
  - 10: Make prediction using Equation (21)
  - 11: **Output** :  $\mu_{t+1}, \hat{r}_t$
- 

### 3.3. Changepoint identification

In addition to providing drug demand forecasting, BOCD also identifies changepoints. At each time  $t$ , we get the Maximum A Posteriori (MAP) over the posterior run length distribution  $P(r_t|x_{1:t})$  in (3) as the changepoint, denoted as  $\hat{r}_t$ , that is,

$$\hat{r}_t = \arg \max_{r_t=0, 1, \dots, t} \frac{P(r_t, x_{1:t})}{\sum_{r_t} P(r_t, x_{1:t})}. \quad (22)$$

If there is no changepoint,  $\hat{r}_t$  should grow consecutively as  $t$  increases. If  $\hat{r}_t$  suddenly drops, it indicates a potential changepoint. However, an outlier may also cause a temporary drop, after which  $\hat{r}_t$  will return to its original scale quickly. To accurately identify real changepoints, we declare a changepoint only if  $\hat{r}_t$  does not return to its original scale for at least ten time steps.

### 3.4. Extension to multiple drug demand forecasting and changepoint detection

This section generalizes the analysis for multivariate data, which could be beneficial when multiple drug demands are highly correlated. We begin by reformulating the problem.

Let  $x^{(i)}$  be a random variable representing the  $i$ th drug demand. Suppose there are  $D$  drug demands  $x^{(1)}, \dots, x^{(D)}$  and we have observed  $D$  data sequences up to time  $t$ , denoted as  $x_{1:t}^{(1)}, \dots, x_{1:t}^{(D)}$ . We use  $x_{1:t}^{(-i)}$  to denote the  $D-1$  sequences except the  $i$ th drug demand  $x_{1:t}^{(i)}$ . Our goal is to predict future data  $x_{t+1}^{(i)}$ , for  $i = 1, \dots, D$ , for each data stream by leveraging information from the  $D$  historical data sequences. Consequently, the changepoint probability for each data sequence is evaluated based on its correlation with the other  $D-1$  data sequences. Let  $r_t^{(i)}$  represent the run length for variable  $x^{(i)}$  of the  $i$ th drug at time  $t$  for  $\forall i = 1, \dots, D$ . Utilizing the information from  $D$  drug demand sequences, we evaluate the posterior marginal run length probability by

$$P(r_t^{(i)} | x_{1:t}^{(i)}, \dots, x_{1:t}^{(D)}) = \frac{P(r_t^{(i)}, x_{1:t}^{(i)} | x_{1:t}^{(-i)})}{P(x_{1:t}^{(i)} | x_{1:t}^{(-i)})}, \quad (23)$$

Using the technique similar to that in deriving (1), we write the conditional joint run length in a recursive manner as follows.

$$P(r_t^{(i)}, x_{1:t}^{(i)} | x_{1:t}^{(-i)}) = \sum_{r_{t-1}^{(i)}} P(r_t^{(i)} | r_{t-1}^{(i)}) P(x_t^{(i)} | r_{t-1}^{(i)}, x_{1:t-1}^{(i)}, x_{1:t}^{(-i)}) P(r_{t-1}^{(i)}, x_{1:t-1}^{(i)} | x_{1:t-1}^{(-i)}) \quad (24)$$

Compared to the analysis for a single drug, the key difference is that we now use a conditional predictive probability  $P(x_t^{(i)} | r_{t-1}^{(i)}, x_{1:t-1}^{(i)}, x_{1:t}^{(-i)})$  in the second term, which incorporates information from other drugs. For the recursion, assuming  $P(r_{t-1}^{(i)}, x_{1:t-1}^{(i)} | x_{1:t-1}^{(-i)}) = P(r_{t-1}^{(i)}, x_{1:t-1}^{(i)} | x_{1:t-1}^{(-i)})$  in the last term, we get

$$P(r_t^{(i)}, x_{1:t}^{(i)} | x_{1:t}^{(-i)}) = \sum_{r_{t-1}^{(i)}} P(r_t^{(i)} | r_{t-1}^{(i)}) P(x_t^{(i)} | r_{t-1}^{(i)}, x_{1:t-1}^{(i)}, x_{1:t}^{(-i)}) P(r_{t-1}^{(i)}, x_{1:t-1}^{(i)} | x_{1:t-1}^{(-i)}) \quad (25)$$

Next, we derive the closed-form expression for  $P(x_t^{(i)} | r_{t-1}^{(i)}, x_{1:t-1}^{(i)}, x_{1:t}^{(-i)})$  under a multivariate Gaussian distribution. This method can also be extended to other types of distributions using Bayesian techniques. Let  $X = [x^{(1)} \dots x^{(D)}]^T$  denote a multivariate random vector that follows a multivariate Gaussian distribution with known covariance, i.e.,  $X \sim \mathcal{N}_D(\theta, \Sigma)$ , where  $\theta$  is a  $D \times 1$  vector representing the mean parameter and  $\Sigma$  is a  $D \times D$  covariance matrix. We use  $X_t$  to represent the realization of  $X$  at time  $t$  and  $X_{1:t}$  to denote the observed data matrix  $X_{1:t} = [x_{1:t}^{(1)} \dots x_{1:t}^{(D)}]^T$ . The sample mean for  $X_{1:t}$  is written as

$$\bar{X}_{1:t} = \begin{bmatrix} \bar{x}_{1:t}^{(1)} \\ \vdots \\ \bar{x}_{1:t}^{(D)} \end{bmatrix} = \begin{bmatrix} \frac{1}{t} \sum_{i=1}^t x_i^{(1)} \\ \vdots \\ \frac{1}{t} \sum_{i=1}^t x_i^{(D)} \end{bmatrix} \quad (26)$$

We further assume that the mean parameter  $\theta$  obeys another multivariate Gaussian with prior mean  $\theta_0$  and prior covariance  $\Sigma_0$ . Following the Bayes' rule, the posterior distribution for  $\theta$  and the posterior predictive distribution are also multivariate Gaussian as follows.

$$p(\theta | X_{1:t}, \Sigma) = \mathcal{N}_D(\theta | \theta_t, \Sigma_t) \quad (27)$$

$$p(X_{t+1} | X_{1:t}) = \mathcal{N}_D(X_{t+1} | \theta_t, \tilde{\Sigma}_t) \quad (28)$$

where  $\theta_t = \Sigma_t(t\Sigma^{-1}\bar{X}_{1:t} + \Sigma_0^{-1}\mu_0)$ ,  $\Sigma_t = (t\Sigma^{-1} + \Sigma_0^{-1})^{-1}$  and  $\tilde{\Sigma}_t = \Sigma_t + \Sigma$ .

Let  $\theta_t^{(i)}$  denote the  $i$ th element of the vector  $\theta_t$ , and  $\theta_t^{(-i)}$  the vector obtained by removing  $\theta_t^{(i)}$  from  $\theta_t$ . Additionally, let  $\tilde{\Sigma}_t^{(i,-i)}$  denote a  $1 \times (D-1)$  submatrix of  $\tilde{\Sigma}_t$ , which includes the  $i$ th row and all columns except the  $i$ th column. Similarly,  $\tilde{\Sigma}_t^{(-i,-i)}$  is a  $(D-1) \times (D-1)$  submatrix of  $\tilde{\Sigma}_t$ , including all rows and columns except those corresponding to the  $i$ th element.

The conditional predictive distribution for  $x_{t+1}^{(i)}$  is a univariate normal distribution described by

$$p(x_{t+1}^{(i)} | X_{1:t}, x_{t+1}^{(-i)}) = \mathcal{N}(x_{t+1}^{(i)} | \theta_t^{(i)}, \tilde{\Sigma}_t^{(i,-i)}), \quad (29)$$

with

$$\theta_t^{(i,-i)} = \theta_t^{(i)} + \tilde{\Sigma}_t^{(i,-i)} (\tilde{\Sigma}_t^{(-i,-i)})^{-1} (\theta_t^{(-i)} - \theta_t^{(-i)}) \quad (30)$$

and

$$\tilde{\Sigma}_t^{(i,-i)} = \tilde{\Sigma}_t^{(i,i)} - \tilde{\Sigma}_t^{(i,-i)} (\tilde{\Sigma}_t^{(-i,-i)})^{-1} (\tilde{\Sigma}_t^{(-i,-i)})^T, \quad (31)$$

for  $i = 1, \dots, D$ .

Likewise, it holds that  $P(x_t^{(i)} | r_{t-1}^{(i)}, x_{1:t-1}^{(i)}, x_{1:t}^{(-i)}) = \mathcal{N}(x_t^{(i)} | \theta_{t-1,r}^{(i,-i)}, \tilde{\Sigma}_{t-1,r}^{(i,-i)})$  where  $\theta_{t-1,r}^{(i,-i)}$  and  $\tilde{\Sigma}_{t-1,r}^{(i,-i)}$  are the run-length dependent statistics calculated using observations only from time  $t-r-1$  onwards, rather than from the beginning. This is because BOCD assumes that once a changepoint occurs, the data sequence shifts to a different distribution with parameters that are different from the previous distribution and thus, data observed before a changepoint do not provide information that is useful for inferring future data points.

Similar to the univariate analysis, we can update the hyperparameters  $\theta_t$  and  $\Sigma_t$  sequentially using the following rules:

$$\Sigma_{t+1} = (\Sigma_t^{-1} + \Sigma^{-1})^{-1} \quad (32)$$

$$\theta_{t+1} = \Sigma_{t+1}(\Sigma_t^{-1}\theta_t + \Sigma^{-1}X_{t+1}) \quad (33)$$

The detailed derivation is available in the [Appendix](#). Regarding the computational complexity, we need to compute the  $D \times D$  precision matrices  $\Sigma_t^{-1}$  and  $\Sigma^{-1}$  at each time  $t$ , which requires  $\mathcal{O}(D^3)$  time complexity. By applying Sherman-Morrison-Woodbury formula (Sherman & Morrison, 1949), we can reduce the cost to  $\mathcal{O}(D^2)$ . Therefore, the total computational cost of the multivariate analysis is  $\mathcal{O}(D^2T^2)$ , where  $T$  denotes the number of observations accumulated up to the current time, and  $D$  denote the number of drugs included in the analysis.

#### 4. Numerical results

This section presents the numerical results obtained from applying our modeling framework presented in [Section 3](#) to a real life dataset from our partnering hospital's central pharmacy. We first introduce the dataset in [Section 4.1](#). Subsequently, we evaluate the model's performance in



Section 4.2 and Section 4.3. In Section 4.4, we present the results of extending our analysis to multivariate drug demand data. Moreover, we demonstrate the practical implications of using our model on inventory costs in Section 5. Finally, our observation summary is presented in Section 4.5.

#### 4.1. Data

The data considered in this study consist of time series data on the daily demand for drug types stored at the partnering hospital's central pharmacy from January 11<sup>th</sup>, 2021 to August 27<sup>th</sup>, 2023. We selected the top 500 most used drugs in this time window. After removing drugs with incomplete data, our dataset consists of 446 drug types. For each drug type, we also know its price per unit. The median and average drug price are \$0.55 and \$49.17 respectively, indicating a skewed drug price distribution. Indeed, the 95th percentile of the drug prices is \$59.22. Table 2 summarizes the statistics of the drug demand data over the three years (2021–2023). We use the weekly demand and normalize each drug demand using min-max scaling. The dataset is split into training and test sets. Specifically, we use the data from 2021 as the training set (51 points) and the data from 2022 to 2023 as the testing sets (88 points). Note that due to the limited amount of available data and the infrequent occurrence of changepoints in the hospital pharmacy setting, we use more testing data than training data. This allows us to observe changepoints in the testing data and better demonstrate the model's prediction performance under changepoint scenarios.

We observe the demand patterns of 111 drug types that have experienced changepoints, which constitute 25.9% of all drug types during the test periods; see Table 3. Here, the changepoints are detected by BOCD using Algorithm 1, and false alarms are removed based on the identification rule in Section 3.3. Some drugs were observed to have up to three changepoints in their demand trajectories. This table also presents information about the average timing of the changepoints. We observe that for drugs with one

changepoint, the first changepoint happens after about 1 year. For drugs with more than one changepoint, the first changepoint seems to happen much earlier, about after approximately half a year.

Interestingly, there seems to be a positive correlation between the number of changepoints and the price. Figure 2 depicts the drug price distributions for drugs with 0, 1, 2, and 3 changepoints. We observe that the median drug prices in each box plot increase with the number of changepoints. It indicates that more expensive drugs tend to experience abrupt demand changes more frequently.

Furthermore, Table 4 summarizes the number of changepoints in each price range. Expensive drugs within the third and fourth quantile price range are almost three times more likely to have changepoints compared to inexpensive drugs in the first quantile. This trend underscores the importance of incorporating potential changepoint occurrences into drug demand prediction.

#### 4.2. Comparison with alternative approaches

To assess the performance of the BOCD model, we compare with other benchmark prediction approaches. We consider several traditional statistical models, including the persistent model, MA, ARIMA, and SARIMA. We also consider Long Short-Term Memory (LSTM), as well as the standard Bayesian model. Specifically, we use a moving average model of order 2, denoted by MA2. For ARIMA and SARIMA models, we use the training dataset to select the best model order based on the Akaike Information Criterion (AIC) for each drug type. For the LSTM model, we set the hyperparameters as follows: the number of memory units is 20, the time step is 15, and the number of epochs is 20. These values were selected as the optimal hyperparameters through cross-validation on the training set of all drug types. To

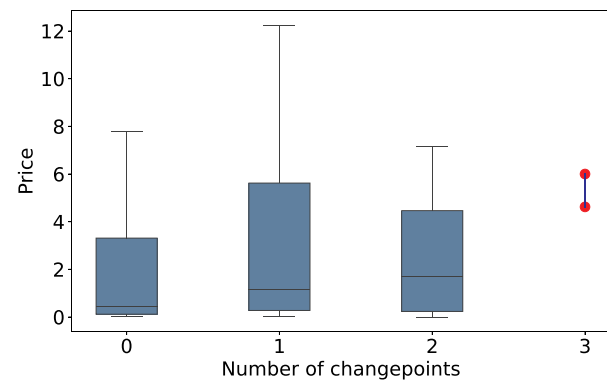


Figure 2. Box plots of the drug price per number of changepoints.

Table 2. Summary statistics of drug demand data considered in this study.

| Summary statistics          | Value              |
|-----------------------------|--------------------|
| Number of drugs             | 446 types          |
| Mean demand                 | 452.45 units/week  |
| Standard deviation demand   | 1176.76 units/week |
| Average price               | \$49.17            |
| Median price                | \$0.55             |
| Standard deviation of price | \$364.67           |

Table 3. Summary statistics of drug demand data in the testing set.

| Summary statistics        | Number of drugs | Time of first changepoint <sup>a</sup> | Duration between changepoints (weeks). |
|---------------------------|-----------------|----------------------------------------|----------------------------------------|
| Drugs with 0 changepoints | 335             |                                        |                                        |
| Drugs with 1 changepoint  | 91              | 52.2                                   |                                        |
| Drugs with 2 changepoints | 18              | 30.6                                   | 27.7                                   |
| Drugs with 3 changepoints | 2               | 21.0                                   | 25.5                                   |

<sup>a</sup>Note that all changepoints are detected using BOCD.

Table 4. Number of drugs that experience abrupt demand changes in the testing set in each price bracket.

| Quantile | Price quantile range | Number of drugs | Drugs with changepoints |
|----------|----------------------|-----------------|-------------------------|
| 25%      | \$0 – \$0.14         | 111             | 14                      |
| 50%      | \$0.15 – \$0.55      | 109             | 23                      |
| 75%      | \$0.56 – \$4.45      | 110             | 39                      |
| 100%     | \$4.45 – \$5261.64   | 111             | 33                      |

adapt to online prediction, the coefficients for ARIMA, SARIMA and LSTM are also updated each time a new datum is observed. Recall that we use a small amount of training data at the beginning of the process, the online update also ensures more accurate parameter estimation as more data becomes available over time.

For the BOCD model, we use the sample mean and sample variance of the training data set for  $\mu_0$ . Similarly, we set  $\sigma_0^2$  and  $\sigma_x^2$  to be the sample variance of the training data. Since we do not expect the changepoints to occur frequently, we set  $\lambda = 100$ , which suggests a changepoint is expected to occur every 100 wk, or roughly every two years.

In our experiments, the seasonal effects in the SARIMA model are found to be insignificant across all the drug types considered in this study. Therefore, we exclude SARIMA from further analysis.

To demonstrate the predictive capabilities of each model in comparison to alternative approaches, Fig. 3 presents the boxplots of RMSE values for all drug types across models. The LSTM model exhibits a relatively large inter-quantile range, suggesting that its performance is sensitive to the drug type. It has been known that LSTM typically performs well with large datasets. Thus, the LSTM's mean RMSE may decrease with larger historical drug demand data, however, the heterogeneity among drugs is likely to persist due to inherent differences in their demand patterns. The BOCD and ARIMA models exhibit the best performance,

characterized by both a low median and a narrow interquartile range. Therefore, we focus on these two models in the subsequent discussion for a more in-depth analysis.

### 4.3. Prediction performance

#### 4.3.1. Evaluation on point prediction

Let us compare the prediction accuracy for the two drugs depicted in Fig. 1. Figure 4 displays the results on the test data for these two drugs. The changepoints of demand trajectories, identified by the proposed BOCD approach, are depicted using the dashed red vertical line at  $t = 53$  for drug ID 003404 and  $t = 63$  for drug ID 003207. The red trajectories depict the predictions from BOCD. Clearly, after BOCD identifies these change points, it quickly adapts its predictions to the shifted process. The mean demand is changed from  $\mu = 0.44$  to  $0.12$  in the first case, while in the second case, the mean shifts from  $\mu = 0.33$  to  $0.64$ . Note that there are some time lags to identify changepoints in both cases, which are unavoidable in order to retrain the model. If such time lags are not allowed, false alarms would frequently occur with noisy demand data.

For the sake of illustration, we also include the predictions from the traditional Bayesian prediction model, indicated by the yellow prediction trajectories in Fig. 4. While it provides online sequential learning and updates the predictive distribution with each new data point, it does not account for changepoints. As a result, it adapts to the changes slowly. Moreover, as can be concluded from the green prediction trajectories in Fig. 4, the ARIMA approach performs better than the standard Bayesian approach; however, it too fails to respond to sudden demand changes.

Next, we evaluate the overall Root Mean Squared Error (RMSE) performance between BOCD and ARIMA. Specifically, we compute the RMSE difference for each drug by subtracting the RMSE of ARIMA from that of BOCD. Fig. 5 presents the distribution of RMSE differences in the testing set across all drugs. Bins on the negative x-axis represent drugs for which BOCD outperforms ARIMA, while those on the positive x-axis indicate drugs where ARIMA outperforms BOCD. While the RMSE difference is close to zero for many drugs, the overall distribution is left-skewed with a long left tail.

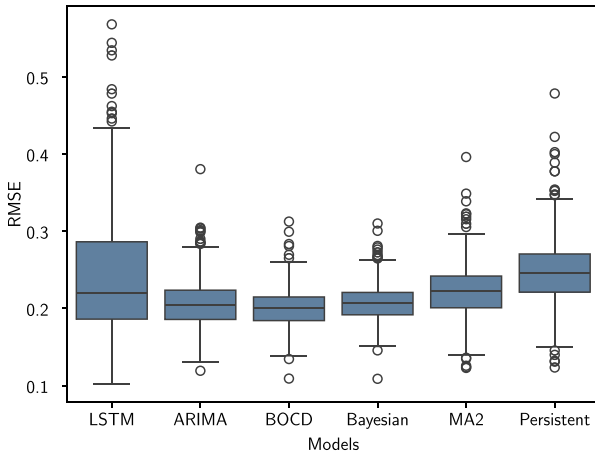


Figure 3. Boxplots of RMSE in each approach in the testing set.

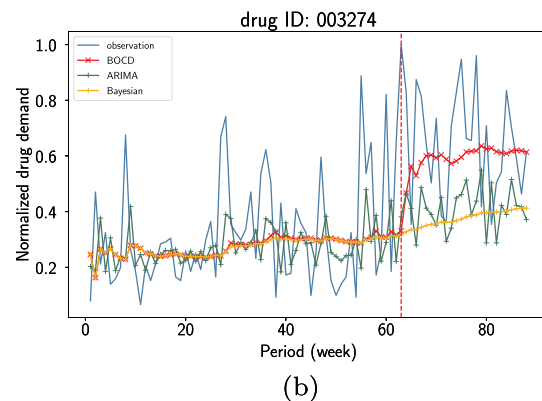
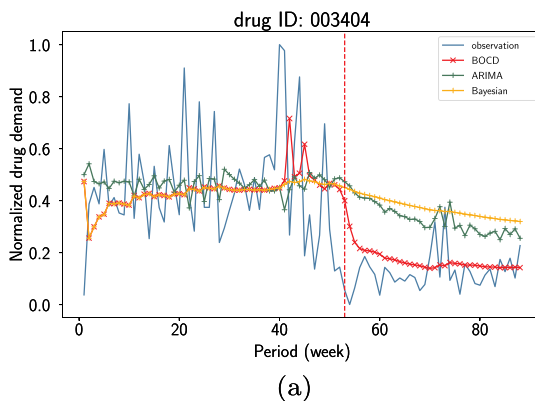


Figure 4. Examples of prediction results for drugs with changepoints.

To assess whether the RMSE difference is substantial enough to conclude that one model outperforms the other, we calculate the sample standard deviation of these differences, denoted as  $\sigma_d$ . Among the 446 drugs included in this study, 75 exhibit a significant performance difference between BOCD and ARIMA, where the difference exceeds  $\sigma_d = 0.021$ . Of the 75 drugs, BOCD outperforms ARIMA in 60 cases, whereas ARIMA outperforms BOCD in 15 cases. Notably, when BOCD achieves a smaller overall RMSE than ARIMA (as shown on the negative x-axis), the distribution exhibits a long tail, indicating several instances of large RMSE from ARIMA. Specifically, when focusing on differences exceeding  $2\sigma_d$ , there are 20 drugs where ARIMA shows pronounced underperformance, compared to only one case where BOCD underperforms. These findings suggest that BOCD demonstrates more robust overall performance than ARIMA.

Additionally, we compare the after-changepoint RMSE, defined as the RMSE of 10 consecutive points immediately following a detected changepoint, to demonstrate the robustness of BOCD when a changepoint occurs. There are 111 drugs with changepoints, and we conduct an analysis similar to the comparison with the overall RMSE. Figure 6 presents histograms of the after-changepoint RMSE differences across the 111 drugs with changepoints, with the sample standard deviation, denoted by  $\sigma'_d$ , being 0.057. Out of the 111 drugs with changepoints, 26 shows significant differences, larger than  $\sigma'_d$ , in after-changepoint performance between BOCD

and ARIMA. Specifically, BOCD outperforms ARIMA in 25 of these 26 cases, while ARIMA outperforms in only 1 case. This suggests that BOCD provides substantially better after-changepoint performance.

Similar to Fig. 5, the after-changepoint RMSE difference distribution also exhibits a long left tail, with many drugs showing large RMSE values from ARIMA. Notably, BOCD outperforms in all 6 drugs that fall beyond the  $2\sigma_d$  threshold. In contrast, when ARIMA outperforms BOCD, the RMSE differences are generally small, with no drug exceeding the  $2\sigma'_d$  threshold. These results suggest the superior adaptiveness of BOCD against abrupt changes.

#### 4.3.2. Uncertainty quantification with prediction intervals

One key feature of BOCD is that it provides a full predictive distribution, enabling us to quantify its prediction uncertainties using prediction intervals. Figure 7 shows the 95% prediction intervals of BOCD and ARIMA for the two example drugs. The intervals are wide due to large fluctuations in the data, but we observe that the intervals of BOCD follow the trend of the data pattern after a changepoint occurs, whereas ARIMA's intervals remain similar throughout the process. This demonstrates that the BOCD model learns changes in the data more quickly than ARIMA.

We additionally investigate the probabilistic forecasting capability with two metrics, including the coverage rate and average width of prediction intervals (Jang et al., 2023). The coverage rate refers to the proportion of observed data that falls within the prediction intervals. A good coverage rate should be close to the nominal rate of the prediction intervals, e.g., 95% when 95% prediction intervals are considered. The average width is the mean difference between the upper and lower bounds of the prediction intervals over the entire test period.

Figure 8 presents boxplots of the coverage rate and average width across all drug types with changepoints using BOCD and ARIMA models. Figure 8a shows BOCD generally provides a higher coverage rate than ARIMA, close to the 95% nominal rate dictated by the red dashed line. Typically, higher average coverage rates are obtained with wider prediction intervals. However, Fig. 8b shows that the width of BOCD's prediction intervals are similar to that of ARIMA's. In fact, BOCD's prediction intervals have a slightly narrower average width of 0.649 compared to ARIMA's 0.704. These results indicate that BOCD offers an enhanced performance in probabilistic forecasting and uncertainty quantification compared to ARIMA.

#### 4.3.3. Relationship with price

Finally, we assess the correlation between the difference in drug demand prediction errors and the price. Figure 9 contains scatter plots along with heat maps. Figure 9a demonstrates that the distribution of the overall RMSE difference between the two methods in the testing set is centered around 0, regardless of the price. On the contrary, we observe the skewed distribution in Fig. 9b when we consider the prediction during the first ten weeks after a changepoint.

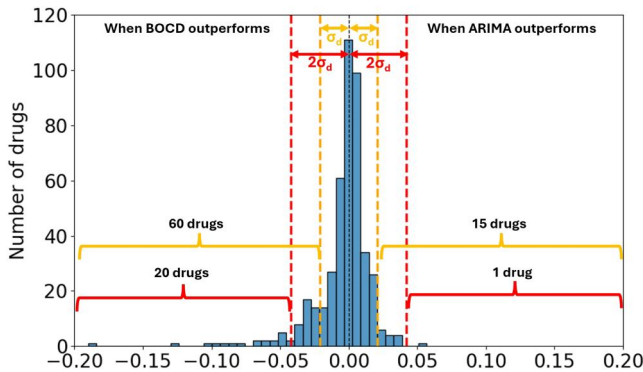


Figure 5. Histogram of overall RMSE difference (BOCD - ARIMA) in the testing set.

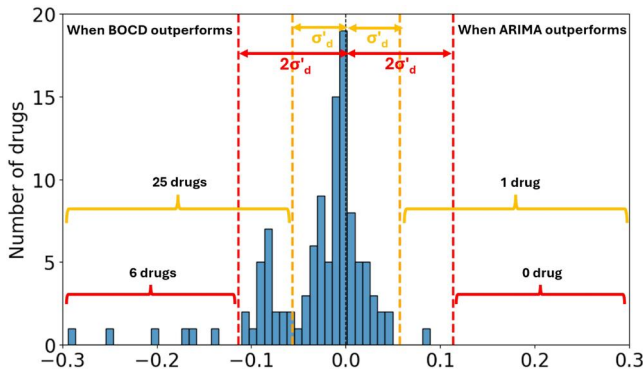


Figure 6. Histogram of after-changepoint RMSE difference (BOCD - ARIMA) in the testing set.

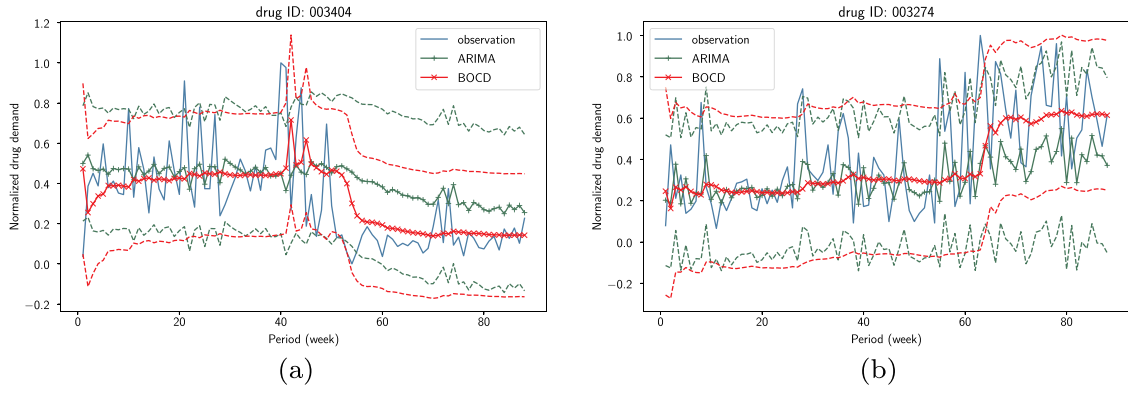


Figure 7. 95% prediction intervals for two drugs with changepoints.

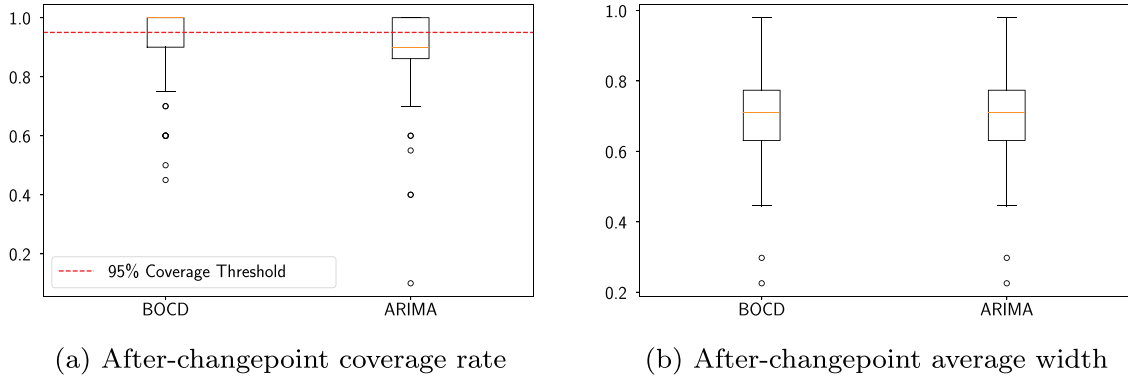


Figure 8. Uncertainty quantification with 95% prediction intervals from BOCD and ARIMA in the testing set.

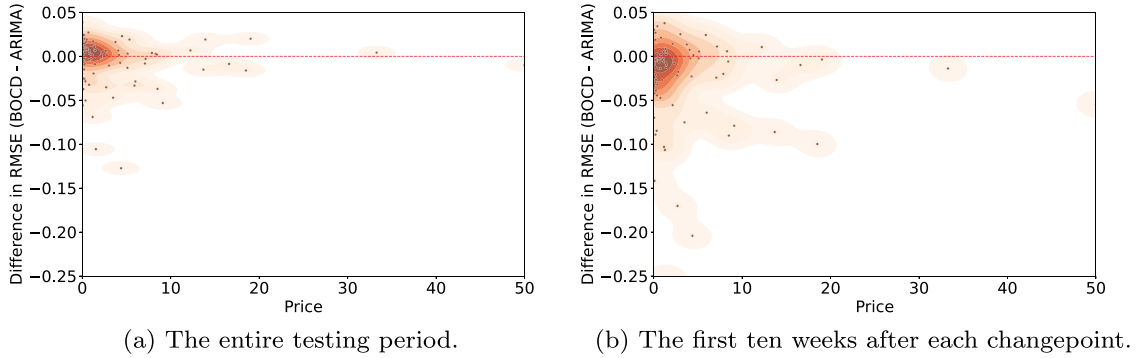


Figure 9. Heat maps showing the relation between the difference in RMSE (BOCD - ARIMA) and the price in the testing set.

Similar to our previous discussion, BOCD produces smaller RMSEs across different price ranges when changepoint occurs. In particular, this benefit is more clear for high-priced drugs. For the drugs whose price is over \$15 (10.9%), BOCD generates more accurate results in most cases.

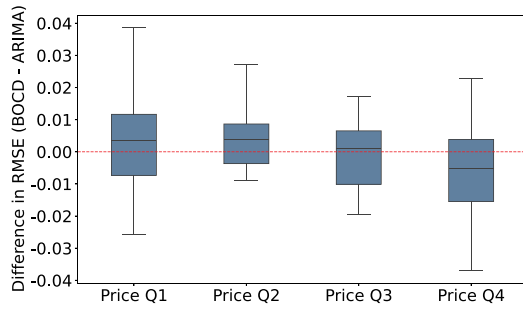
To further investigate, Fig. 10 reports the distribution of the RMSE difference in different price ranges for the overall test set and after changepoint occurrences. In both cases, there seem to be some correlation between the price and the RMSE difference; BOCD tends to produce smaller RMSEs for more expensive drugs. This trend is more evident in Fig. 10b when we compare the RMSEs after the changepoints. Considering that the high-priced drugs experience more abrupt changes in drug demands, as we observe in Section 4.1, these results imply that BOCD is more beneficial in forecasting demands for relatively higher priced drugs.

#### 4.4. Performances of multivariate BOCD

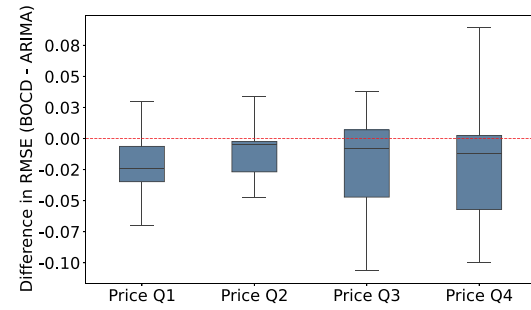
In this section, we evaluate the performance of multivariate BOCD when we analyze multiple drug demands together. To illustrate the approach, we choose two drugs, as shown in Fig. 11, where both drugs appear to have a changepoint at around  $t = 30$ .

Figure 12 compares the posterior run length distributions for drug ID #001609 under both univariate and multivariate BOCD analysis. The shading represents the probability density of the run length at each time point, with darker colors indicating higher densities. At each time step, the run length with the highest probability density is highlighted in red. In the absence of a changepoint, we expect to see a continuous straight red line; when a changepoint or a false alarm occurs, the red line drops to zero and begins anew. False

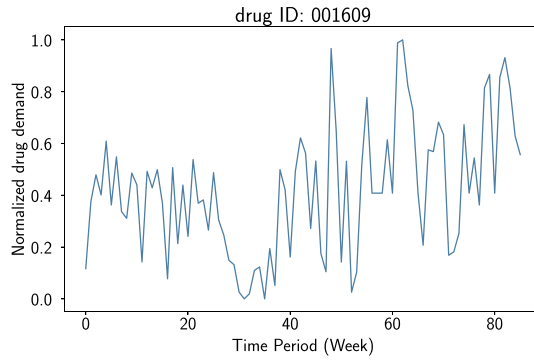




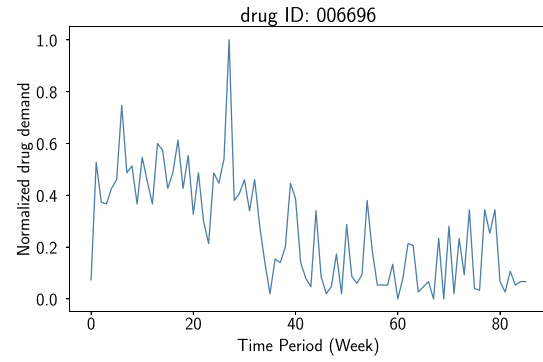
(a) The entire testing period.



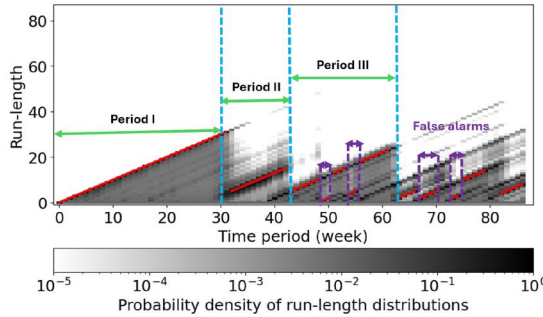
(b) The first ten weeks after each changepoint.

**Figure 10.** RMSE difference (BOCD-ARIMA) vs. drug price in the testing set.

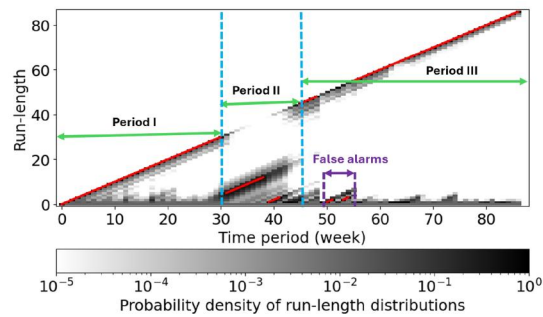
(a)



(b)

**Figure 11.** Examples of two drugs with similar changepoint locations.

(a) Univariate BOCD



(b) Multivariate BOCD

**Figure 12.** Posterior run length distribution for drug ID #001609.

alarm periods are indicated by purple intervals. We observe that the changepoints identified by the univariate and multivariate BOCD are consistent. However, the univariate BOCD produces more false alarms during Period III compared to the multivariate BOCD. This indicates that the multivariate BOCD can reduce the uncertainty of changepoint location by leveraging information from other drugs.

Figure 13 presents the posterior run length distributions for drug ID #006696. In both univariate and multivariate BOCD analyses, the changepoint is clearly identified without any false alarms. However, during Period I, the univariate BOCD, shown in Fig. 13a, exhibits non-zero densities across all possible run lengths, as indicated by the relatively darker color under the red line. In contrast, the multivariate BOCD in Fig. 13b concentrates the density at a single obvious run

length, with most other run lengths having zero probability, as indicated by the lighter color under the red line. Our analysis demonstrates that multivariate analysis reduces the uncertainty in the changepoint distribution when multiple drug demands are highly correlated.

In summary, we demonstrate that the multivariate approach can make more accurate decisions regarding changepoint detection by incorporating additional information from other drugs. In particular, the decrease in false alarms (i.e., less false positives) for changepoint can be very useful in practice by surfacing actual changes to operational personnel, thereby improving their workflows. Future research could focus on multiple highly correlated drugs to uncover more meaningful interpretations, such as those in a similar pharmaceutical category.



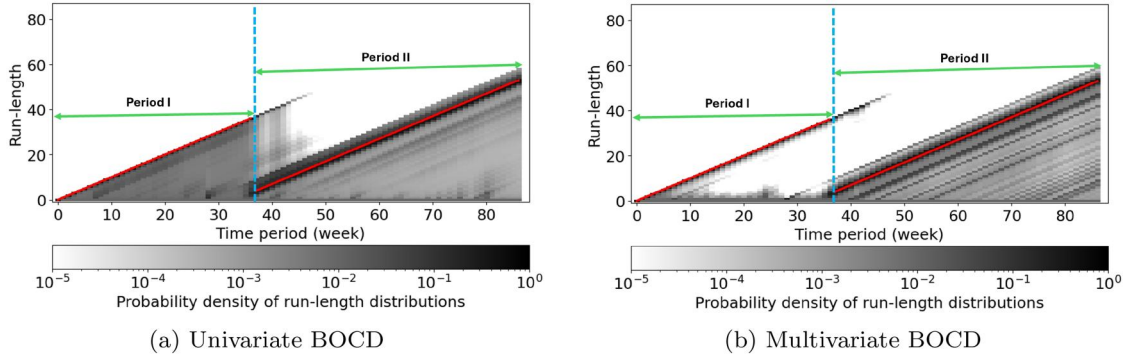


Figure 13. Posterior run length distribution for drug ID #006696.

#### 4.5. Summary

Our analysis reveals that 25.9% of all drugs exhibit at least one significant changepoint in their demand trajectory over the observed period. This finding underscores the dynamic nature of drug demand, where abrupt shifts can occur, potentially driven by external factors such as regulatory changes, market entry of competitors, or shifts in prescribing practices. When examining the relationship between these changepoints and drug pricing, we observed a striking trend: about 66% of these changepoints occur within the upper half of the price spectrum. This suggests that higher-priced drugs may be more susceptible to fluctuations in demand, possibly due to factors such as market sensitivity to pricing strategies or the introduction of generic alternatives.

In terms of point forecasting accuracy, our comparative analysis of BOCD and the ARIMA model reveals that, overall, there are no significant differences in their performance across the entire dataset. Despite this general trend, a more granular examination focusing on the periods immediately following a changepoint presents a different narrative. Furthermore, BOCD demonstrates more robust performance than ARIMA in both overall RMSE and after-changepoint RMSE. It produces significantly smaller prediction errors in cases where it outperforms, while only slightly increased errors in cases where it underperforms. In terms of probabilistic forecasting, BOCD outperforms ARIMA by achieving a higher coverage rate and a narrower average prediction width.

The convergence of these findings leads to an important insight: for drugs that fall within the higher price range, employing BOCD as a forecasting method is particularly advantageous. The ability of BOCD to quickly adapt to changepoints offers a significant edge in accurately predicting future demand in the aftermath of these shifts. This suggests that for more expensive drugs, where the stakes of forecasting errors are higher, leveraging BOCD could lead to better-informed decision-making, thereby optimizing inventory management and pricing strategies. Additionally, multivariate BOCD shows potential for improving the changepoint detection capability by accounting for the correlations among multiple drug demands.

#### 5. Application to inventory management

One of the primary uses of forecasting methods in the supply chain context is to optimize inventory levels, helping to prevent both over- and understocking of drugs, thereby improving fiscal accountability and quality of care, respectively. Sudden disruptions in demand patterns can lead to severe drug shortages due to abrupt demand increases or significant wastage if demand decreases. Timely detection of changepoints and adjusting forecasts accordingly can mitigate these risks, ensuring better inventory management.

To demonstrate the benefits of accurate forecasting when the underlying drug demand process changes, we employ a simple Newsvendor model (Silver et al., 2017, Chapter 9). This well-studied model is used to optimize order quantities. Let  $c_o$  and  $c_u$  denote overstocking and understocking costs per each week for one drug unit, respectively. Assuming that the lead time is negligible, the optimal order quantity  $q_t$  can be obtained as

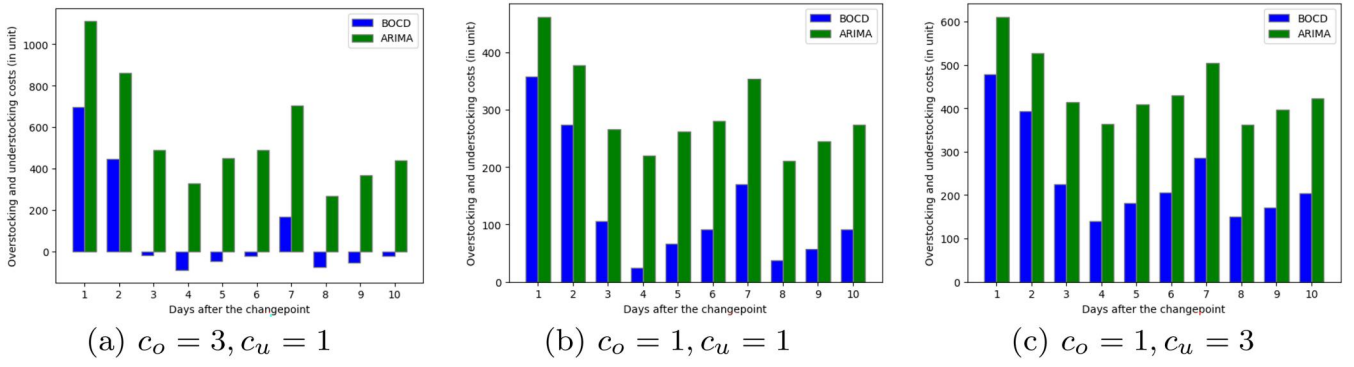
$$q_t = F_t^{-1} \left( \frac{c_u}{c_o + c_u} \right), \quad (34)$$

where  $F_t^{-1}$  is the inverse of the cumulative distribution function of the demand at time  $t$ . In other words,  $q_t$  is the  $c_u/(c_o + c_u) \times 100$  percentile of the demand distribution (Wang et al., 2021).

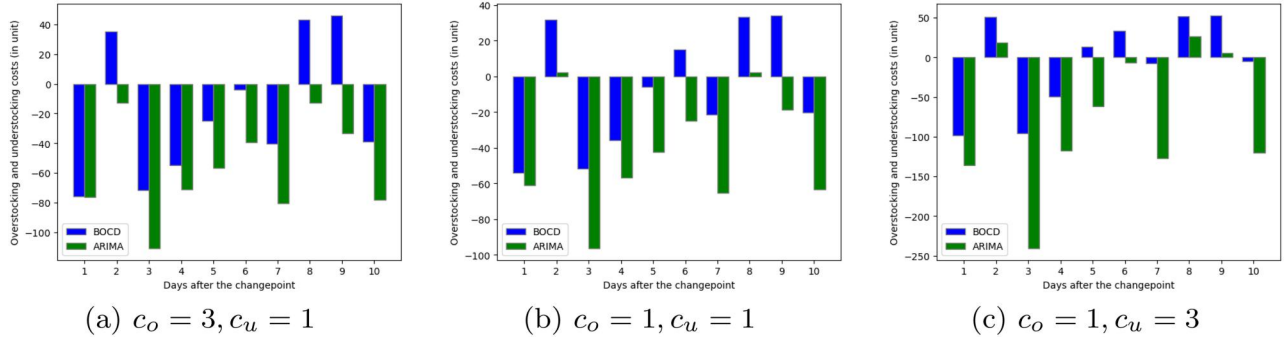
Assuming backlog is allowed, this order quantity can be adjusted based on the current backlog or inventory level. Let  $b_{t-1}$  and  $h_{t-1}$  represent the backlog and inventory at time  $t-1$ , respectively. The adjusted order quantity can then be calculated to ensure optimal alignment with current stock levels as follows.

$$q_t^* = (q_t + b_{t-1}) \cdot I(b_{t-1} > 0) + \max(q_t - h_{t-1}, 0) \cdot I(h_{t-1} > 0). \quad (35)$$

The first term addresses the demand at time  $t$  when there is a backlog at time  $t-1$ , whereas the second term considers the stock available at time  $t-1$ . After the true demand at time  $t$  is realized, the total overstocking and understocking costs at time  $t$  becomes  $c_u \cdot b_t \cdot I(b_t > 0) + c_o \cdot h_t \cdot I(h_t > 0)$  with  $b_t = \max(x_t - q_t^* - h_{t-1} + b_{t-1}, 0)$  and  $h_t = \max(q_t^* + h_{t-1} - b_{t-1} - x_t, 0)$ .



**Figure 14.** Overstocking and understocking costs for drug ID #003404 (note: Positive values in the y-axis imply the costs due to overstocking, while negative values represent the costs associated with understocking.).



**Figure 15.** Overstocking and understocking costs for drug ID #003274. (note: Positive values in the y-axis imply the costs due to overstocking, while negative values represent the costs associated with understocking.).

Using the predictive distribution  $F_t$  obtained from the BOCD method, we compute the over- and understocking costs for drugs that experience abrupt demand changes. We focus on the two drugs illustrated in Fig. 4. Recall that drug ID #003404 experiences a sudden decrease in mean demand, while drug ID #003274 experiences a sudden increase in mean demand.

As the actual overstocking and understocking costs for these drugs are not available, we consider various hypothetical scenarios for  $c_o$  and  $c_u$ . The first scenario penalizes overstocking more than understocking, meaning  $c_o/c_u > 1$ . For illustration purpose, we set  $c_o = 3$  and  $c_u = 1$ , treating these costs unit-less. Under these cost settings,  $q_t$  in (35) corresponds to the 25th percentile of the predictive demand distribution. In the second scenario, we consider overstocking and understocking as equally undesirable with  $c_o = 1$  and  $c_u = 1$ , resulting in  $q_t$  in (35) being the median of the predictive demand distribution. Lastly, the third scenario penalizes understocking more than overstocking, with  $c_o/c_u < 1$ , so  $q_t$  should be greater than the median. For this scenario, we set  $c_o = 1$  and  $c_u = 3$ , making  $q_t$  the 75th percentile of the predictive demand distribution.

We compute the overstocking and understocking costs for the first ten weeks following a changepoint and compare the costs from both ARIMA and BOCD at each time point. The weekly overstocking and understocking costs for drug ID #003404 and drug ID #003274 across three different scenarios are shown in Figs. 14 and 15, respectively. In these figures, a positive value in the y-axis indicates overstocking, while a negative value implies understocking. The height of

the bars represents the magnitude of the associated overstocking or understocking costs.

For drug ID #003404, where the mean demand decreases after the changepoint, overstocking occurs more likely. Notably, BOCD results in lower overstocking costs than ARIMA across all three scenarios. In particular, as shown in Fig. 14a, where overstocking costs are higher than understocking costs, the advantage of BOCD is more pronounced. In this scenario, BOCD results in some weeks of understocking, but the backlog amount is minimal. Conversely, ARIMA consistently incurs substantial overstocking over time. This highlights BOCD's strong advantage over ARIMA in reducing drug wastage, especially when there is a sudden decrease in drug demand and overstocking costs are high. In the other two scenarios with different cost settings, BOCD still consistently results in lower overstocking costs.

Similarly, Fig. 15 present the overstocking and understocking costs for drug ID #003274. As the mean demand of this drug increases after the changepoint, understocking occurs more likely. In all scenarios, the understocking costs of BOCD are consistently smaller than those of ARIMAs. Although BOCD sometimes leads to overstocking in certain weeks, likely due to high data variability, it generally demonstrates superiority over ARIMA in managing drug shortages, particularly when there is a sudden increase in demand. The advantage of BOCD is clear in Fig. 15c when  $c_u$  is higher than  $c_o$ , making understocking a more significant concern. In such scenarios, BOCD effectively mitigates the risks associated with insufficient inventory levels.

**Table 5.** Weekly average costs after the changepoint for drug ID #003404.

| Scenario           | Method | Overstocking cost | Understocking cost | Total cost |
|--------------------|--------|-------------------|--------------------|------------|
| $c_o = 3, c_u = 1$ | BOCD   | 130.77            | 33.58              | 164.35     |
|                    | ARIMA  | 550.71            | 0                  | 550.71     |
| $c_o = 1, c_u = 1$ | BOCD   | 127.38            | 0                  | 127.38     |
|                    | ARIMA  | 294.91            | 0                  | 294.91     |
| $c_o = 1, c_u = 3$ | BOCD   | 243.59            | 0                  | 243.59     |
|                    | ARIMA  | 443.97            | 0                  | 443.97     |

**Table 6.** Weekly average costs after the changepoint for drug ID #003274.

| Scenario           | Method | Overstocking cost | Understocking cost | Total cost |
|--------------------|--------|-------------------|--------------------|------------|
| $c_o = 3, c_u = 1$ | BOCD   | 12.46             | 31.07              | 43.53      |
|                    | ARIMA  | 0                 | 57.20              | 57.20      |
| $c_o = 1, c_u = 1$ | BOCD   | 11.38             | 18.94              | 30.32      |
|                    | ARIMA  | 0.43              | 42.91              | 43.34      |
| $c_o = 1, c_u = 3$ | BOCD   | 20.30             | 25.67              | 45.96      |
|                    | ARIMA  | 5.05              | 81.07              | 86.12      |

Tables 5 and 6 present the average weekly overstocking and understocking costs, as well as the total weekly average costs under different scenarios for both drugs. These results are consistent with the observations from Figs. 14 and 15. Across all cases, BOCD generates smaller total costs, offering savings of 45% to 70% savings for drug ID #003404 and 24% to 47% savings for drug ID #003274, respectively.

## 6. Conclusion

In this paper, we propose the use of BOCD models to cope with changepoints in drug demand. In our case study, 25.9% of included drugs have demand changepoints. Over 65% of drugs with changepoints are in the upper half of the price range for all included drugs. This underpins the need for specialized methods that are able to quickly recognize and adapt to changepoints, to prevent misalignment of drug inventories. We have shown that, especially right after the changepoint happened, BOCD outperforms conventional methods, thanks to its ability to quickly adapt to changepoints. The method is particularly promising for drugs in the upper half of the price range, as these drugs tend to have more change points, and rapid adaptation is of utmost importance to prevent misaligned inventory decisions.

Additionally, we extend the analysis from a single-drug setting to a multi-drug context by generalizing the BOCD method from a univariate to a multivariate framework. By leveraging information across multiple drugs, the method reduces the uncertainty associated with changepoint detection. This improvement is particularly beneficial when strong correlations exist among the drugs. Finally, we demonstrate the effectiveness of BOCD in managing inventory costs. Specifically, BOCD reduces overstocking costs when demand suddenly decreases and mitigates understocking costs when demand abruptly increases.

One of the limitations of our study is that our data only has a single factor, being the drug's price to compare forecasting performance. We hypothesize that, if available, additional factors such as main drug use, the surgical schedule and patient characteristics was available, thus could lead to more accurate predictions and useful implications. For instance, using these variables we could cluster the drugs,

and develop dedicated models per cluster. The additional information would also help in explaining why certain models outperform others. We also suspect that an even longer time series could be beneficial.

The results obtained are based on the hospital pharmacy setting. Although our model is generic in nature, it would be of great interest to see how our model and its results generalize across different types of drugs or markets. Drugs with different demand patterns, regulatory environments, or market dynamics may require tailored models, and the effectiveness of BOCD in these contexts remains untested. This might be an interesting avenue for further research. Future studies could also delve deeper into understanding the causal mechanisms behind demand changepoints. For example, by identifying the underlying reasons for changepoints, whether due to changes in medical guidelines, patient demographics, or external market factors, more tailored interventions could be developed.

We have shown that changepoints models such as BOCD are able to predict and react to shifts in patterns in a pharmaceutical context, without compromising predictive performance with respect to traditional methods. Due to the flexibility of the provided BOCD model, it would be of great interest to investigate whether we can improve its performance using more exogenous variables. If important factors affecting drug demand are known, our approach can be extended to include linear regression models. Furthermore, the impact of a changepoint may vary depending on the availability of substitutes and whether the drug is life-critical, which represents an important direction for future work.

We applied our model to a subset of 446 drugs from the extensive range available at our partner hospital, demonstrating the effectiveness of our methods and the advantages they offer. Exploring further scalability, we're also excited to assess how well the model performs as we expand our dataset. While the BOCD method may require additional computational resources for large datasets or when incorporating multiple external variables, this exploration could offer valuable insights. Another idea would be to look into how to use the demand predictions in diverse inventory management contexts. Investigating how BOCD predictions could be integrated into other inventory optimization strategies beyond the newsvendor model could be highly valuable. This might involve exploring how different inventory policies (e.g., just-in-time, safety stock) respond to changepoint predictions and how to optimize them accordingly.

In conclusion, our model offers crucial insights for hospital pharmacy decision-makers by highlighting drugs that are likely to experience shifts in demand in the near future. While there are avenues to further refine the model introduced in this paper, we believe this work holds significant value for any healthcare system where effective demand forecasting is essential.

## Acknowledgement(s)

The authors thank our partnering hospital for providing us with data and background information.

## Disclosure statement

No conflicts of interest were reported by the author(s).

## Funding

This research was funded by Diakonessenhuis Utrecht, the Netherlands, and the Bernhard Cultuurfonds.

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## Appendix

This section provides derivations for Equations (27) through (31) in Section 3.4. We use the same notation as shown in Table 1. Suppose  $X_k \sim \mathcal{N}_D(\theta, \Sigma)$ ,  $1 \leq k \leq t$ . When no changepoint occurs between time 1 to time  $t$ , the likelihood of observing  $X_{1:t}$  is given by:

$$\begin{aligned} L(X_{1:t}|\theta, \Sigma) &= \prod_{k=1}^t f(X_k|\theta, \Sigma) \\ &= (2\pi)^{-D/2} \det(\Sigma)^{-t/2} \exp \left\{ -\frac{1}{2} \sum_{k=1}^t (X_k - \theta)^T \Sigma^{-1} (X_k - \theta) \right\}. \end{aligned}$$

Additionally, we assume the covariance parameter  $\Sigma$  is known and further assume that the mean parameter  $\theta$  follows another multivariate normal distribution with mean  $\theta_0$  and covariance  $\Sigma_0$ , i.e.,  $\theta \sim \mathcal{N}_D(\theta_0, \Sigma_0)$ . Applying Bayes' rule, we obtain the posterior distribution for  $\theta$  as

$$\begin{aligned} p(\theta|X_{1:t}) &\propto L(X_{1:t}|\theta, \Sigma) \times p(\theta|\theta_0, \Sigma_0) \\ &\propto \exp \left\{ -\frac{1}{2} \sum_{k=1}^t (X_k - \theta)^T \Sigma^{-1} (X_k - \theta) \right\} \exp \left\{ -\frac{1}{2} (\theta - \theta_0)^T \Sigma_0^{-1} (\theta - \theta_0) \right\} \\ &\propto \exp \left\{ -\frac{1}{2} \left( \sum_{k=1}^t X_k^T \Sigma^{-1} X_k + t \theta^T \Sigma^{-1} \theta - 2t \theta^T \Sigma^{-1} \bar{X}_{1:t} \right. \right. \\ &\quad \left. \left. + \theta^T \Sigma_0^{-1} \theta + \theta_0^T \Sigma_0^{-1} \theta_0 - 2\theta^T \Sigma_0^{-1} \theta_0 \right) \right\} \\ &\propto \exp \left\{ -\frac{1}{2} \left[ \theta^T (t \Sigma^{-1} + \Sigma_0^{-1}) \theta - 2\theta^T (t \Sigma^{-1} \bar{X}_{1:t} + \Sigma_0^{-1} \theta_0) \right] \right\} \\ &\propto \exp \left\{ -\frac{1}{2} (\theta - \theta_t)^T \Sigma_t^{-1} (\theta - \theta_t) \right\} \end{aligned}$$

where  $\theta_t = \Sigma_t (t \Sigma^{-1} \bar{X}_{1:t} + \Sigma_0^{-1} \theta_0)$  and  $\Sigma_t = (t \Sigma^{-1} + \Sigma_0^{-1})^{-1}$ . Next, we derive the posterior predictive distribution  $p(X_{t+1}|X_{1:t})$ :

$$\begin{aligned} p(X_{t+1}|X_{1:t}) &= \int_{\theta \in \mathbb{R}^D} p(X_{t+1}|\theta, \Sigma) p(\theta|X_{1:t}) d\theta \\ &= \int_{\theta \in \mathbb{R}^D} \mathcal{N}_D(X_{t+1}; \theta, \Sigma) \mathcal{N}_D(\theta; \theta_t, \Sigma_t) d\theta \\ &= \int_{\theta \in \mathbb{R}^D} \mathcal{N}_D(X_{t+1} - \theta; \mathbf{0}, \Sigma) \mathcal{N}_D(\theta; \theta_t, \Sigma_t) d\theta \end{aligned}$$

Therefore, the posterior distribution of  $X_{t+1}$  is the distribution of the sum of the independent random variables  $X_{t+1} - \theta$  and  $\theta$ , which is also a multivariate normal distribution with mean  $\theta_t$  and covariance  $\Sigma + \Sigma_t$ . This completes the derivation for (27) and (28). Additionally, when a changepoint occurs, we use the data from the most recent changepoint. For example, suppose the most recent changepoint occurs at time  $r$ , we replace  $X_{1:t}$  with  $X_{(t-r+1):t}$  in the above equations.

To ensure the time efficiency of online learning, we derive the sequential update rules of  $\theta_t$  and  $\Sigma_t$  presented in (32) and (33). Let  $\Lambda_t = \Sigma_t^{-1}$ , we have

$$\begin{aligned} \Lambda_{t+1} &= \Sigma_{t+1}^{-1} \\ &= (t+1) \Sigma^{-1} + \Sigma_0^{-1} \\ &= (t \Sigma^{-1} + \Sigma_0^{-1}) + \Sigma^{-1} \\ &= \Lambda_t + \Sigma^{-1} \end{aligned} \quad (36)$$

Consequently,  $\Sigma_{t+1} = \Lambda_t^{-1} = (\Sigma_t^{-1} + \Sigma^{-1})^{-1}$ . Furthermore, we rewrite  $\theta_{t+1}$  as follows.

$$\begin{aligned} \theta_{t+1} &= \Sigma_{t+1} ((t+1) \Sigma^{-1} \bar{X}_{1:t+1} + \Sigma_0^{-1} \mu_0) \\ &= \Sigma_{t+1} \left( \Sigma^{-1} \sum_{k=1}^{t+1} X_k + \Sigma_0^{-1} \mu_0 \right) \\ &= \Sigma_{t+1} \left( \Sigma^{-1} \left( \sum_{k=1}^t X_k + X_{t+1} \right) + \Sigma_0^{-1} \mu_0 \right) \\ &= \Sigma_{t+1} \left( \Sigma^{-1} \sum_{k=1}^t X_k + \Sigma^{-1} X_{t+1} + \Sigma_0^{-1} \mu_0 \right) \\ &= \Sigma_{t+1} ((t \Sigma^{-1} \bar{X}_{1:t} + \Sigma_0^{-1} \mu_0) + \Sigma^{-1} X_{t+1}) \\ &= \Sigma_{t+1} (\theta_t + \Sigma^{-1} X_{t+1}). \end{aligned}$$

Next, we derive the conditional predictive distribution for the  $i$ -th drug, given the past information of all drugs. Specifically, we aim to obtain  $p(x_{t+1}^{(i)}|x_{t+1}^{(-i)}, X_{1:t})$ . Let  $\tilde{\Sigma}_t = \Sigma_t + \Sigma$ . We first rewrite the full predictive distribution in matrix form as

$$X_{t+1}|X_{1:t} = \begin{bmatrix} x_{t+1}^{(1)} \\ \vdots \\ x_{t+1}^{(D)} \end{bmatrix} \Bigg| X_{1:t} \sim \mathcal{N}_D \left( \begin{bmatrix} \theta_t^{(1)} \\ \vdots \\ \theta_t^{(D)} \end{bmatrix}, \begin{bmatrix} \tilde{\Sigma}_t^{(1,1)} & \cdots & \tilde{\Sigma}_t^{(1,D)} \\ \vdots & \ddots & \vdots \\ \tilde{\Sigma}_t^{(D,1)} & \cdots & \tilde{\Sigma}_t^{(D,D)} \end{bmatrix} \right).$$

We split the matrix  $X_{t+1}$  into two parts: a scalar  $x_{t+1}^{(i)}$  and a  $D \times 1$  vector  $x_{t+1}^{(-i)}$ . Consequently, we divide the mean and covariance of the full predictive distribution into two parts. Specifically, for the mean parameter  $\theta$ , let  $\theta_t^{(i)}$  denote the  $i$ th element of the vector  $\theta_t$ , and  $\theta_t^{(-i)}$  the vector obtained by removing  $\theta_t^{(i)}$  from  $\theta_t$ . For the covariance matrix  $\tilde{\Sigma}_t$ , let  $\tilde{\Sigma}_t^{(i,-i)}$  denote the  $1 \times (D-1)$  submatrix of  $\tilde{\Sigma}_t$



corresponding to the  $i$ th row and all columns except the  $i$ th column. Similarly, let  $\tilde{\Sigma}_t^{(-i,-i)}$  denote the  $(D-1) \times (D-1)$  submatrix obtained by removing the  $i$ -th row and column. We can then rewrite the predictive distribution as:

$$X_{t+1}|X_{1:t} = \begin{bmatrix} x_{t+1}^{(i)} \\ x_{t+1}^{(-i)} \end{bmatrix} \Bigg| X_{1:t} \sim \mathcal{N}_D \left( \begin{bmatrix} \theta_t^{(i)} \\ \theta_t^{(-i)} \end{bmatrix}, \begin{bmatrix} \tilde{\Sigma}_t^{(i,i)} & \tilde{\Sigma}_t^{(i,-i)} \\ \tilde{\Sigma}_t^{(-i,i)} & \tilde{\Sigma}_t^{(-i,-i)} \end{bmatrix} \right).$$

By the property of multivariate normal distribution (Eaton, 1983), the conditional distribution of  $x_{t+1}^{(i)}$  given  $x_{t+1}^{(-i)}$  and  $X_{1:t}$  is given by:

where

$$x_{t+1}^{(i)}|x_{t+1}^{(-i)}, X_{1:t} \sim \mathcal{N}_D(\theta_t^{(i|-i)}, \tilde{\Sigma}_t^{(i|-i)})$$

and

$$\theta_t^{(i|-i)} = \theta_t^{(i)} + \tilde{\Sigma}_t^{(i,-i)} (\tilde{\Sigma}_t^{(-i,-i)})^{-1} (x_t^{(-i)} - \theta_t^{(-i)})$$

$$\tilde{\Sigma}_t^{(i|-i)} = \tilde{\Sigma}_t^{(i,i)} - \tilde{\Sigma}_t^{(i,-i)} (\tilde{\Sigma}_t^{(-i,-i)})^{-1} (\tilde{\Sigma}_t^{(-i,i)})^T,$$

for  $i = 1, \dots, D$ .